Stochastic Calculus & Option Pricing — Synthesis (Weeks 1–7)

Nathan Sauldubois

October 8, 2026

How to use these notes (Weeks 1–7).

- These are *synthetic* lecture notes: definitions, key results, and the computations you actually use.
- Each lecture ends with a short checklist: "what you must be able to do".
- Notation: all processes are defined on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$ satisfying the usual conditions.

Contents

1 Lecture 1 — Probability refresher and core mathematical tools **2**
1.1 Random variables, laws, CDFs, and densities . . . 2
1.2 Expectation, moments, variance, and covariance . . . 4
1.3 Sample moments, LLN, and CLT . . . 4
1.4 Characteristic functions . . . 5
1.5 Information, measurability, and conditional expectation . . . 6
1.6 Processes, filtrations, and martingales . . . 8
1.7 Differential-calculus review . . . 8

2 Lecture 2 — Brownian Motion, Quadratic Variation, and GBM **9**
2.1 From Discrete Random Walks to Brownian Motion . . . 9
2.2 Definition and Basic Properties of Brownian Motion . . . 10
2.3 Markov and Strong Markov Properties . . . 10
2.4 Reflection Principle and Hitting Times . . . 11
2.5 Pathwise Regularity and Irregularity . . . 11
2.6 Quadratic Variation . . . 11
2.7 Heuristic Itô Rules . . . 12
2.8 Geometric Brownian Motion (GBM) . . . 12
2.9 Stopping Times and Optional Stopping . . . 12
2.10 Multidimensional Brownian Motion . . . 13

3 Lecture 3 — The Itô Stochastic Integral **14**
3.1 Definition, admissibility, and construction . . . 14
3.2 The identities that do the calculations . . . 15
3.3 Deterministic integrands and formulas to know . . . 16

4 Lecture 4 — Itô Processes and Itô's Formula **17**
4.1 Quadratic covariation and the differential table . . . 17
4.2 One- and two-dimensional Itô formulas . . . 18
4.3 Reusable transformation rules . . . 18
4.4 Canonical SDEs to solve and reuse . . . 19
4.5 Several factors and instantaneous hedging . . . 19

5 Lecture 5 — Self-Financing, Replication, and Black–Scholes **20**
5.1 From Itô's formula to a pricing equation . . . 20
5.2 The self-financing budget . . . 20
5.3 Replication, the PDE, and verification . . . 21
5.4 Gaussian solution, call formula, and Greeks . . . 21

6 Lecture 6 — Reweighting, Girsanov, and Risk-Neutral Measure **22**
6.1 Rare tails motivate a change of weights . . . 22
6.2 Densities and conditional Bayes' formula . . . 22
6.3 Stochastic exponential and the sign of Girsanov . . . 23
6.4 The Black–Scholes pricing measure . . . 23

7 Lecture 7 — Martingale Prices, Hedges, and Completeness **23**
7.1 From stochastic integrals to martingale prices . . . 24
7.2 Dividend gains and the pricing equation . . . 24
7.3 Change of numeraire . . . 24
7.4 Martingale representation turns prices into hedges . . . 25
7.5 Completeness and the spanning condition . . . 25
7.6 Path dependence and the choice of state . . . 25

1 Lecture 1 — Probability refresher and core mathematical tools

Summary. Random variables and their laws; CDFs and densities; moments, variance, and covariance; dominated convergence; LLN and CLT; characteristic functions; information and conditional expectation; Gaussian conditioning; filtrations and martingales; and the differential-calculus formulas used later.

Reference. S. E. Shreve, *Stochastic Calculus for Finance II*, Chapters 1–2.

1.1 Random variables, laws, CDFs, and densities

Random variable and law. On a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, a real random variable is a measurable map $X : (\Omega, \mathcal{F}) \to (\mathbb{R}, \mathcal{B}(\mathbb{R}))$. Its law is the probability measure

$$\mathbb{P}_X(A) = \mathbb{P}(X \in A), \qquad A \in \mathcal{B}(\mathbb{R}).$$

The random variable maps states to numbers; its law records the probabilities of those numbers.

CDF and density. The cumulative distribution function of X is

$$F_X(x) = \mathbb{P}(X \le x).$$

Every CDF is nondecreasing and right-continuous, with limits 0 at $-\infty$ and 1 at $+\infty$. For $a < b$,

$$\mathbb{P}(a < X \le b) = F_X(b) - F_X(a).$$

If X has density f_X, then

$$F_X(x) = \int_{-\infty}^{x} f_X(u)\,\mathrm{d}u, \qquad \mathbb{P}(a < X \le b) = \int_a^b f_X(u)\,\mathrm{d}u,$$

and $f_X = F_X'$ wherever the CDF is differentiable. A law and a CDF always exist; a density need not exist.

Joint and marginal laws. The joint law of (X, Y) contains both marginal laws and their dependence. In the discrete case,

$$p_X(x) = \sum_y p_{X,Y}(x, y), \qquad p_Y(y) = \sum_x p_{X,Y}(x, y),$$

whereas for a joint density,

$$f_X(x) = \int_{\mathbb{R}} f_{X,Y}(x, y)\,\mathrm{d}y, \qquad f_Y(y) = \int_{\mathbb{R}} f_{X,Y}(x, y)\,\mathrm{d}x.$$

Independence is equivalent to factorization of the joint law: $p_{X,Y} = p_X p_Y$ or $f_{X,Y} = f_X f_Y$.

Common one-dimensional laws.

Law	Parameters	$\mathbb{E}[X]$	$\mathrm{Var}(X)$
Bernoulli	p	p	$p(1-p)$
Binomial	n, p	np	$np(1-p)$
Poisson	λ	λ	λ
Uniform	$[a, b]$	$(a+b)/2$	$(b-a)^2/12$
Exponential	rate λ	$1/\lambda$	$1/\lambda^2$
Gaussian	μ, σ^2	μ	σ^2

Mass functions and densities to recognize. For the discrete laws,

$$\begin{aligned}
X \sim \mathrm{Bernoulli}(p): &\quad \mathbb{P}(X = x) = p^x(1-p)^{1-x}, \quad x \in \{0, 1\},\\
X \sim \mathrm{Binomial}(n, p): &\quad \mathbb{P}(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0, \dots, n,\\
X \sim \mathrm{Poisson}(\lambda): &\quad \mathbb{P}(X = k) = e^{-\lambda}\frac{\lambda^k}{k!}, \quad k \in \mathbb{N}.
\end{aligned}$$

For the continuous laws,

$$\begin{aligned}
X \sim \mathrm{Unif}[a, b]: &\quad f_X(x) = \frac{1}{b-a}\mathbf{1}_{[a,b]}(x),\\
X \sim \mathrm{Exp}(\lambda): &\quad f_X(x) = \lambda e^{-\lambda x}\mathbf{1}_{[0,\infty)}(x),\\
X \sim \mathcal{N}(\mu, \sigma^2): &\quad f_X(x) = \frac{1}{\sigma\sqrt{2\pi}}\exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right).
\end{aligned}$$

1.2 Expectation, moments, variance, and covariance

Expectation of a transformation. If the relevant sum or integral is finite, then

$$\mathbb{E}[h(X)] = \sum_x h(x)\mathbb{P}(X = x) \quad \text{or} \quad \mathbb{E}[h(X)] = \int_{\mathbb{R}} h(x) f_X(x)\,\mathrm{d}x.$$

The kth moment is $\mathbb{E}[X^k]$. Integrability means $\mathbb{E}[|X|] < \infty$, and $\mathbb{E}[\mathbf{1}_A] = \mathbb{P}(A)$.

Essential expectation properties. For integrable X, Y and constants a, b,

$$\begin{aligned} \mathbb{E}[aX + bY] &= a\mathbb{E}[X] + b\mathbb{E}[Y], \\ X \le Y &\Longrightarrow \mathbb{E}[X] \le \mathbb{E}[Y], \\ |\mathbb{E}[X]| &\le \mathbb{E}[|X|]. \end{aligned}$$

Linearity does not require independence.

Variance and covariance. For square-integrable variables,

$$\mathrm{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2] - \mathbb{E}[X]^2, \qquad \mathrm{SD}(X) = \sqrt{\mathrm{Var}(X)},$$

and

$$\mathrm{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y].$$

The identities used most often are

$$\begin{aligned} \mathrm{Var}(aX + b) &= a^2\mathrm{Var}(X), \\ \mathrm{Var}(X + Y) &= \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\mathrm{Cov}(X, Y), \\ \mathrm{Var}\left(\sum_{k=1}^n X_k\right) &= \sum_{k=1}^n \mathrm{Var}(X_k) + 2\sum_{i<j} \mathrm{Cov}(X_i, X_j). \end{aligned}$$

Independence implies zero covariance, but the converse is false in general. For independent square-integrable variables, the variance of a sum is the sum of the variances.

Theorem 1.1 (Dominated Convergence Theorem)**.** *Suppose $X_n \to X$ almost surely and there is an integrable random variable Y such that $|X_n| \le Y$ almost surely for every n. Then X is integrable and*

$$\mathbb{E}[|X_n - X|] \longrightarrow 0, \qquad \mathbb{E}[X_n] \longrightarrow \mathbb{E}[X].$$

Remark 1.2. Dominated convergence is the standard tool for interchanging a limit and an expectation. Pointwise convergence alone is not sufficient; the integrable dominating variable controls the tails uniformly in n.

1.3 Sample moments, LLN, and CLT

For observations $X_1, \ldots, X_n$, define

$$\overline{X}_n = \frac{1}{n}\sum_{i=1}^n X_i, \qquad v_n = \frac{1}{n}\sum_{i=1}^n (X_i - \overline{X}_n)^2, \qquad s_n^2 = \frac{n}{n-1} v_n.$$

Here v_n is the empirical variance and s_n^2 is its usual unbiased version under i.i.d. sampling with finite variance.

Theorem 1.3 (Law of Large Numbers)**.** *If (X_i) are i.i.d. and $\mathbb{E}[|X_1|] < \infty$, then*

$$\overline{X}_n \xrightarrow[n\to\infty]{\mathbb{P}} \mu = \mathbb{E}[X_1].$$

Theorem 1.4 (Central Limit Theorem)**.** *If, in addition, $0 < \sigma^2 = \mathrm{Var}(X_1) < \infty$, then*

$$\frac{\sqrt{n}(\overline{X}_n - \mu)}{\sigma} \Rightarrow \mathcal{N}(0,1).$$

Thus $\overline{X}_n$ has approximate standard deviation $\sigma/\sqrt{n}$ for large n.

The LLN says that the estimation error vanishes; the CLT gives its $n^{-1/2}$ scale and approximate Gaussian shape.

1.4 Characteristic functions

Definition and basic properties. The characteristic function of X is

$$\varphi_X(u) = \mathbb{E}[e^{iuX}], \qquad u \in \mathbb{R}.$$

It always exists because $|e^{iuX}| = 1$. Moreover,

$$\varphi_X(0) = 1, \qquad |\varphi_X(u)| \le 1, \qquad \varphi_{aX+b}(u) = e^{iub}\varphi_X(au).$$

The characteristic function uniquely determines the law. If $\mathbb{E}[|X|^k] < \infty$, then it can be differentiated at the origin and

$$\varphi_X^{(k)}(0) = i^k \mathbb{E}[X^k].$$

Independent sums. If X and Y are independent, then

$$\varphi_{X+Y}(u) = \varphi_X(u)\varphi_Y(u).$$

This turns convolution of laws into multiplication of functions.

Formulas to know.

$$\begin{aligned}
X \sim \mathrm{Bernoulli}(p): &\quad \varphi_X(u) = 1 - p + pe^{iu}, \\
X \sim \mathrm{Poisson}(\lambda): &\quad \varphi_X(u) = \exp\big(\lambda(e^{iu} - 1)\big), \\
X \sim \mathrm{Exp}(\lambda): &\quad \varphi_X(u) = \frac{\lambda}{\lambda - iu}, \\
X \sim \mathcal{N}(\mu, \sigma^2): &\quad \varphi_X(u) = \exp\left(i\mu u - \frac{1}{2}\sigma^2 u^2\right).
\end{aligned}$$

For $Z \sim \mathcal{N}(0,1)$, differentiation under the integral followed by integration by parts gives

$$\varphi_Z'(u) = -u\varphi_Z(u), \qquad \varphi_Z(0) = 1.$$

Hence $\varphi_Z(u) = e^{-u^2/2}$; writing $X = \mu + \sigma Z$ gives the general Gaussian formula.

1.5 Information, measurability, and conditional expectation

σ-algebras as information. A sub-σ-algebra $\mathcal{G} \subseteq \mathcal{F}$ represents the events that can be decided from the available information. A variable Z is $\mathcal{G}$-measurable when its value can be determined from that information. The information revealed by a signal Y is $\sigma(Y)$.

On a finite space, $\mathcal{G}$ is generated by a partition $A_1, \dots, A_m$. A variable is $\mathcal{G}$-measurable if and only if it is constant on every atom A_j. Equivalently,

$$Z \text{ is } \sigma(Y)\text{-measurable} \quad \Longleftrightarrow \quad Z = g(Y)$$

for some measurable g.

Conditional probability and conditional laws. For events A, B with $\mathbb{P}(B) > 0$,

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

For discrete variables and $\mathbb{P}(Y = y) > 0$,

$$\mathbb{P}(X = x \mid Y = y) = \frac{\mathbb{P}(X = x, Y = y)}{\mathbb{P}(Y = y)}.$$

For a joint density and $f_Y(y) > 0$,

$$f_{X|Y}(x \mid y) = \frac{f_{X,Y}(x, y)}{f_Y(y)}, \qquad f_Y(y) = \int_{\mathbb{R}} f_{X,Y}(x, y)\, \mathrm{d}x.$$

Definition of conditional expectation. For $X \in L^1$ and $\mathcal{G} \subseteq \mathcal{F}$, the conditional expectation $\mathbb{E}[X \mid \mathcal{G}]$ is the almost-surely unique random variable Z such that

$$Z \text{ is } \mathcal{G}\text{-measurable}, \qquad \int_A Z \, \mathrm{d}\mathbb{P} = \int_A X \, \mathrm{d}\mathbb{P} \quad \text{for every } A \in \mathcal{G}.$$

It is the prediction of X based only on the information in $\mathcal{G}$. Conditioning on Y means conditioning on $\sigma(Y)$:

$$\mathbb{E}[X \mid Y] = \mathbb{E}[X \mid \sigma(Y)].$$

Finite-partition formula. If $\mathcal{G} = \sigma(A_1, \dots, A_m)$, where the A_j are partition atoms with positive probability, then

$$\mathbb{E}[X \mid \mathcal{G}] = \sum_{j=1}^{m} \frac{\mathbb{E}[X \mathbf{1}_{A_j}]}{\mathbb{P}(A_j)} \mathbf{1}_{A_j} = \sum_{j=1}^{m} \mathbb{E}[X \mid A_j] \mathbf{1}_{A_j}.$$

Conditional expectation is therefore the average of X inside each information set.

Formula for $\mathbb{E}[h(X) \mid Y]$: discrete case. Define, for every y with $\mathbb{P}(Y = y) > 0$,

$$\begin{aligned} m(y) &= \mathbb{E}[h(X) \mid Y = y] \\ &= \sum_x h(x) \mathbb{P}(X = x \mid Y = y) \\ &= \frac{\sum_x h(x) \mathbb{P}(X = x, Y = y)}{\mathbb{P}(Y = y)}. \end{aligned}$$

Then

$$\boxed{\mathbb{E}[h(X) \mid Y] = m(Y).}$$

Formula for $\mathbb{E}[h(X) \mid Y]$: density case. For $f_Y(y) > 0$, define

$$\begin{aligned} m(y) &= \mathbb{E}[h(X) \mid Y = y] \\ &= \int_{\mathbb{R}} h(x) f_{X|Y}(x \mid y)\,\mathrm{d}x \\ &= \frac{\int_{\mathbb{R}} h(x) f_{X,Y}(x,y)\,\mathrm{d}x}{\int_{\mathbb{R}} f_{X,Y}(x,y)\,\mathrm{d}x}. \end{aligned}$$

Again,

$$\boxed{\mathbb{E}[h(X) \mid Y] = m(Y).}$$

The final substitution $y = Y$ is essential: a conditional expectation is a random variable, not merely a number indexed by y.

Essential conditional-expectation properties. For integrable variables, constants a, b, and $\mathcal{H} \subseteq \mathcal{G} \subseteq \mathcal{F}$,

$$\begin{aligned} \mathbb{E}[aX + bY \mid \mathcal{G}] &= a\mathbb{E}[X \mid \mathcal{G}] + b\mathbb{E}[Y \mid \mathcal{G}], \\ X \le Y &\Longrightarrow \mathbb{E}[X \mid \mathcal{G}] \le \mathbb{E}[Y \mid \mathcal{G}], \\ Z\ \mathcal{G}\text{-measurable} &\Longrightarrow \mathbb{E}[ZX \mid \mathcal{G}] = Z\mathbb{E}[X \mid \mathcal{G}], \\ X\ \mathcal{G}\text{-measurable} &\Longrightarrow \mathbb{E}[X \mid \mathcal{G}] = X, \\ X \text{ independent of } \mathcal{G} &\Longrightarrow \mathbb{E}[X \mid \mathcal{G}] = \mathbb{E}[X], \\ \mathbb{E}[\mathbb{E}[X \mid \mathcal{G}] \mid \mathcal{H}] &= \mathbb{E}[X \mid \mathcal{H}]. \end{aligned}$$

In particular,

$$\mathbb{E}[\mathbb{E}[X \mid \mathcal{G}]] = \mathbb{E}[X].$$

For convex ϕ, conditional Jensen gives

$$\phi(\mathbb{E}[X \mid \mathcal{G}]) \le \mathbb{E}[\phi(X) \mid \mathcal{G}]$$

whenever the quantities are integrable.

Gaussian conditioning. If (X, Y) is jointly Gaussian, with standard deviations $\sigma_X, \sigma_Y > 0$ and correlation ρ, then

$$X \mid (Y = y) \sim \mathcal{N}\left(\mu_X + \rho\frac{\sigma_X}{\sigma_Y}(y - \mu_Y),\ \sigma_X^2(1 - \rho^2)\right).$$

Consequently,

$$\mathbb{E}[X \mid Y] = \mu_X + \rho\frac{\sigma_X}{\sigma_Y}(Y - \mu_Y), \qquad \mathrm{Var}(X \mid Y) = \sigma_X^2(1 - \rho^2).$$

The conditional mean is affine in the observed signal, and the residual variance decreases as $|\rho|$ increases.

1.6 Processes, filtrations, and martingales

Processes and filtrations. A discrete-time stochastic process is a sequence $(X_n)_{n\ge 0}$ of random variables on the same probability space. A filtration is an increasing family of σ-algebras,

$$\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \mathcal{F}_2 \subseteq \cdots,$$

representing information accumulated through time. The natural filtration of X is

$$\mathcal{F}_n^X = \sigma(X_0, \dots, X_n).$$

A process is adapted if X_n is $\mathcal{F}_n$-measurable for every n. Adaptedness is the no-look-ahead requirement.

Random walks. If

$$S_n = S_0 + \sum_{k=1}^{n} \xi_k,$$

where the increments are i.i.d. with mean m and variance σ^2, then

$$\mathbb{E}[S_n] = S_0 + nm, \qquad \mathrm{Var}(S_n) = n\sigma^2.$$

Martingales. An integrable adapted process (M_n) is a martingale if

$$\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] = M_n \qquad \text{for every } n.$$

For the random walk above,

$$M_n = S_n - nm$$

is a martingale. For independent symmetric ± 1 increments, $S_n^2 - n$ is also a martingale. A martingale combines three requirements: integrability, adaptation, and the conditional fair-game identity.

1.7 Differential-calculus review

One-variable derivative and Taylor expansion. For $g : \mathbb{R} \to \mathbb{R}$,

$$g'(x) = \lim_{h \to 0} \frac{g(x+h) - g(x)}{h}.$$

If g is twice differentiable near x, then

$$g(x+h) = g(x) + g'(x)h + \frac{1}{2} g''(x) h^2 + o(h^2).$$

Gradient and Hessian. For $f : \mathbb{R}^d \to \mathbb{R}$,

$$\nabla f(x) = \begin{pmatrix} \partial_1 f(x) \\ \vdots \\ \partial_d f(x) \end{pmatrix}, \qquad \nabla^2 f(x) = \big(\partial_{ij} f(x)\big)_{1 \le i,j \le d}.$$

The gradient gives first-order sensitivities; the Hessian records curvature and interactions. The second-order multivariable expansion is

$$f(x+h) = f(x) + \nabla f(x)^\top h + \frac{1}{2} h^\top \nabla^2 f(x) h + o(\|h\|^2).$$

Multivariable chain rule. If $x(t) \in \mathbb{R}^d$ and f are differentiable, then

$$\frac{\mathrm{d}}{\mathrm{d}t} f(x(t)) = \nabla f(x(t))^\top x'(t).$$

In particular, for $V = V(t, s)$ and a differentiable path $S(t)$,

$$\frac{\mathrm{d}}{\mathrm{d}t} V(t, S(t)) = V_t(t, S(t)) + V_s(t, S(t))S'(t).$$

The second-order Taylor term involving V_{ss} will be essential once the underlying path is Brownian rather than differentiable.

Lecture 1 checklist

You should be able to:

- pass between a law, a CDF, a density, and interval probabilities;
- compute $\mathbb{E}[h(X)]$, moments, variances, covariances, and variances of sums;
- state when dominated convergence allows a limit to pass through $\mathbb{E}$;
- distinguish the conclusions of the LLN and the CLT;
- compute and use characteristic functions, especially for independent sums;
- identify the information in $\sigma(Y)$ and test measurability on partition atoms;
- compute $\mathbb{E}[h(X) \mid Y]$ in both discrete and density models;
- use linearity, known factors, independence, monotonicity, Jensen, and the tower rule;
- apply the Gaussian conditioning formula;
- verify adaptation and the martingale property;
- compute derivatives, gradients, Hessians, Taylor expansions, and chain rules.

2 Lecture 2 — Brownian Motion, Quadratic Variation, and GBM

References

- *Stochastic Calculus for Finance II: Continuous-Time Models*, S. E. Shreve, Chapter 3, pp. 83-117
- *Brownian Motion and Stochastic Calculus*, I. Karatzas and S. E. Shreve, Chapter 2, pp. 47-126

2.1 From Discrete Random Walks to Brownian Motion

Simple symmetric random walk. Let $(\varepsilon_k)_{k\geq 1}$ be i.i.d. with $\mathbb{P}(\varepsilon_k = \pm 1) = \frac{1}{2}$, and define

$$S_0 = 0, \qquad S_n = \sum_{k=1}^{n} \varepsilon_k.$$

Then (S_n) is a discrete-time martingale with independent increments, $\mathbb{E}[S_n] = 0$ and $\mathrm{Var}(S_n) = n$.

Diffusive scaling. Introduce a time step $1/n$ and define the rescaled process

$$W^{(n)}(t) = \frac{1}{\sqrt{n}} S_{\lfloor nt \rfloor}, \qquad t \geq 0,$$

with linear interpolation. By the CLT, increments of $W^{(n)}$ over intervals of length $t - s$ are approximately Gaussian with variance $t - s$.

Donsker invariance principle (informal). As $n \to \infty$, $W^{(n)}$ converges in distribution to a continuous-time process W, called *Brownian motion*. This result explains why Brownian motion is the universal continuous-time limit of random walks.

2.2 Definition and Basic Properties of Brownian Motion

Definition. A process $(W_t)_{t \geq 0}$ is a *standard Brownian motion* if:

- $W_0 = 0$,
- it has independent increments,
- for $0 \leq s < t$, $W_t - W_s \sim \mathcal{N}(0, t - s)$,
- paths are continuous almost surely.

Gaussian process viewpoint. Brownian motion is a centered Gaussian process fully characterized by

$$\mathrm{Cov}(W_s, W_t) = \min(s, t).$$

In particular, for $s < t$,

$$W_t = W_s + (W_t - W_s),$$

where W_s and $W_t - W_s$ are independent.

Martingale property. With respect to its natural filtration $\mathcal{F}_t = \sigma(W_s : s \leq t)$, Brownian motion is a martingale:

$$\mathbb{E}[W_t \mid \mathcal{F}_s] = W_s, \qquad s \leq t.$$

2.3 Markov and Strong Markov Properties

Markov property. Brownian motion is Markov: conditionally on W_s, the future is independent of the past, if $t \geq s$,

$$\mathbb{P}(W_t \in A \mid \mathcal{F}_s) = \mathbb{P}(W_t \in A \mid W_s).$$

Equivalently,

$$W_t \mid W_s = x \sim \mathcal{N}(x, t - s).$$

Strong Markov property. Let τ be a stopping time. Conditionally on $\mathcal{F}_\tau$, the shifted process

$$(W_{\tau + t} - W_\tau)_{t \geq 0}$$

is a Brownian motion independent of $\mathcal{F}_\tau$. This is crucial for hitting times, barriers, and American-style features.

2.4 Reflection Principle and Hitting Times

Running maximum. Define the running maximum

$$M_t = \sup_{0 \le s \le t} W_s,$$

and the hitting time of level a:

$$\tau_a = \inf\{t \ge 0 : \ W_t = a\}.$$

Reflection principle. For $a > 0$ and $t > 0$,

$$\mathbb{P}(M_t \ge a) = 2\,\mathbb{P}(W_t \ge a).$$

More generally, for $w \le a$,

$$\mathbb{P}(M_t > a, \ W_t < w) = \mathbb{P}(W_t > 2a - w).$$

Interpretation. Among paths that hit level a before time t, reflecting the path after the first hitting time preserves Wiener measure. This converts path-dependent events into Gaussian tail probabilities.

Consequence. Brownian motion hits every level almost surely:

$$\mathbb{P}(\tau_a < \infty) = 1, \qquad a \in \mathbb{R}.$$

2.5 Pathwise Regularity and Irregularity

Nowhere differentiability. Almost surely, Brownian paths are nowhere differentiable.

Oscillations. For every $t \ge 0$, almost surely,

$$\limsup_{h \downarrow 0} \frac{W_{t+h} - W_t}{\sqrt{h}} = +\infty, \qquad \liminf_{h \downarrow 0} \frac{W_{t+h} - W_t}{\sqrt{h}} = -\infty.$$

Hölder regularity. For every $\alpha < \frac{1}{2}$, Brownian motion is locally α-Hölder, but it is not $\frac{1}{2}$-Hölder.

2.6 Quadratic Variation

Definition. For a partition $\pi = \{0 = t_0 < \cdots < t_n = T\}$, define

$$[X]_T^\pi = \sum_{k=0}^{n-1} (X_{t_{k+1}} - X_{t_k})^2.$$

If this converges as $|\pi| \to 0$, the limit is the quadratic variation $[X]_T$.

Finite variation vs. Brownian motion. If $X_t = \int_0^t v_s ds$, then $[X]_T = 0$. In contrast, for Brownian motion,

$$[W]_T = T.$$

Key martingale. Quadratic variation explains why

$$W_t^2 - t$$

is a martingale. This is the simplest manifestation of Itô calculus.

Quadratic covariation. For processes X, Y,

$$[X, Y]_T = \lim_{\pi} \sum_k (X_{t_{k+1}} - X_{t_k})(Y_{t_{k+1}} - Y_{t_k}).$$

If A has finite variation, then $[A, W] = 0$.

2.7 Heuristic Itô Rules

Brownian increments satisfy the scaling:

$$\Delta W = O(\sqrt{\Delta t}).$$

As a bookkeeping device,

$$(dW_t)^2 = dt, \qquad dW_t\, dt = 0, \qquad (dt)^2 = 0.$$

These rules encode which discrete sums survive in the limit.

2.8 Geometric Brownian Motion (GBM)

Motivation. Additive models allow negative prices and ignore scale effects. Instead, model relative changes:

$$\frac{dS_t}{S_t} = \mu\, dt + \sigma\, dW_t.$$

Explicit solution. The solution is

$$S_t = S_0 \exp\Big((\mu - \tfrac{1}{2}\sigma^2)t + \sigma W_t\Big),$$

so S_t is lognormally distributed.

Moments.

$$\mathbb{E}[S_t] = S_0 e^{\mu t}, \qquad \mathrm{Var}(S_t) = S_0^2 e^{2\mu t}(e^{\sigma^2 t} - 1).$$

Volatility scaling. Log-returns over Δt have variance $\sigma^2 \Delta t$, which justifies the $\sqrt{\Delta t}$ rule.

2.9 Stopping Times and Optional Stopping

Stopped process. For a stopping time τ and a martingale (M_t), define

$$M_t^\tau = M_{t \wedge \tau}.$$

Then (M_t^τ) is again a martingale.

Optional stopping (safe version). If τ is bounded, then

$$\mathbb{E}[M_\tau] = \mathbb{E}[M_0].$$

In general, $\tau < \infty$ a.s. is not sufficient; integrability or uniform integrability is required.

Example. For Brownian motion and $\tau_a = \inf\{t : \ W_t = a\}$, $\mathbb{E}[\tau_a] = \infty$ and $W_{\tau_a} = a$, so $\mathbb{E}[W_{\tau_a}] \neq \mathbb{E}[W_0]$. Optional stopping applies only to $t \wedge \tau_a$.

2.10 Multidimensional Brownian Motion

Definition. A d-dimensional Brownian motion is a vector-valued process

$$W_t = (W_t^1, \dots, W_t^d), \qquad t \geq 0,$$

such that:

- $W_0 = 0 \in \mathbb{R}^d$,
- it has independent increments,
- for $0 \leq s < t$, the increment $W_t - W_s$ is multivariate Gaussian with mean 0 and covariance matrix $(t-s)I_d$,
- paths are continuous almost surely.

Equivalently, the components $(W^1, \dots, W^d)$ are independent one-dimensional Brownian motions.

Gaussian characterization. Multidimensional Brownian motion is a centered Gaussian process with

$$\mathrm{Cov}(W_s^i, W_t^j) = \delta_{ij} \min(s,t), \qquad 1 \leq i,j \leq d,$$

where δ_{ij} is the Kronecker symbol.

Independent vs. correlated Brownian motions. More generally, one may consider a Brownian motion with covariance matrix $\Sigma \in \mathbb{R}^{d\times d}$ (symmetric, positive semidefinite), defined by

$$\mathrm{Cov}(W_s, W_t) = \min(s,t)\,\Sigma.$$

In this case,

$$\mathrm{d}[W^i, W^j]_t = \Sigma_{ij}\ \mathrm{d}t.$$

Any such process can be written as

$$W_t = B\,\widetilde{W}_t,$$

where $\widetilde{W}$ is a standard d-dimensional Brownian motion and B satisfies $BB^\top = \Sigma$ (e.g. Cholesky factorization).

Quadratic variation and covariation. For a d-dimensional Brownian motion W,

$$[W^i]_t = t, \qquad [W^i, W^j]_t = 0 \quad (i \neq j)$$

in the independent case, and more generally

$$[W]_t = t\,\Sigma,$$

where $[W]_t$ denotes the matrix-valued quadratic variation.

Itô rules (multidimensional mnemonic). For independent components,

$$(\mathrm{d}W_t^i)^2 = \mathrm{d}t, \qquad \mathrm{d}W_t^i\ \mathrm{d}W_t^j = 0\ (i \neq j), \qquad \mathrm{d}W_t^i\ \mathrm{d}t = 0, \qquad (\mathrm{d}t)^2 = 0.$$

With correlation matrix Σ, this becomes

$$\mathrm{d}W_t^i\ \mathrm{d}W_t^j = \Sigma_{ij}\ \mathrm{d}t.$$

Preview. In higher dimension, stochastic calculus keeps the same structure: *drift terms*, *martingale terms*, and *quadratic variation*. Only the bookkeeping becomes matrix-valued. This naturally leads to the multidimensional Itô formula and to multidimensional diffusion models.

Lecture 2 checklist

You should be able to:

- state the defining properties of Brownian motion and compute its Gaussian increment laws and covariance function;
- use independent increments to verify the martingale and Markov properties;
- apply the reflection principle to convert a maximum or hitting-time event into a Gaussian tail probability;
- distinguish Brownian paths from finite-variation paths, in particular through their quadratic variation;
- use the Itô multiplication rules $(\mathrm{d}W_t)^2 = \mathrm{d}t$, $\mathrm{d}W_t\,\mathrm{d}t = 0$, and $(\mathrm{d}t)^2 = 0$;
- solve the geometric Brownian-motion SDE and identify the lognormal law of S_t;
- identify a stopping time and state a safe version of optional stopping, including its hypotheses;
- work with independent and correlated multidimensional Brownian motions and their covariations.

3 Lecture 3 — The Itô Stochastic Integral

Summary. The Itô integral is the L^2 limit of left-point, non-anticipative gains. Predictability prevents the use of future information and finite expected energy gives integrability. The resulting gain process is a square-integrable martingale, with quadratic variation $\int_0^t \phi_s^2\,\mathrm{d}s$; for deterministic integrands, L^2 inner products become covariances of jointly Gaussian variables.

3.1 Definition, admissibility, and construction

Why left endpoints are essential. For a grid $0 = t_0 < \cdots < t_n = T$, a discrete gain has the form

$$G_T^\pi = \sum_{i=0}^{n-1} \phi_{t_i}(W_{t_{i+1}} - W_{t_i}).$$

The coefficient used on $(t_i, t_{i+1}]$ must be known at time t_i: ϕ_{t_i} is $\mathcal{F}_{t_i}$-measurable. This is the mathematical content of predictability and the economic content of non-anticipation. The sampling rule matters because Brownian quadratic variation survives:

$$\sum_i W_{t_{i+1}}\Delta W_i - \sum_i W_{t_i}\Delta W_i = \sum_i (\Delta W_i)^2 \longrightarrow T.$$

Definition 3.1 (Simple predictable integrand)**.** A simple predictable process on $[0,T]$ is

$$\phi_t = \sum_{i=0}^{n-1} \xi_i \mathbf{1}_{(t_i,t_{i+1}]}(t), \qquad \xi_i \in \mathcal{F}_{t_i}, \qquad \mathbb{E}[\xi_i^2] < \infty.$$

Its Itô integral is the left-point gain

$$\boxed{\int_0^T \phi_t \, \mathrm{d}W_t = \sum_{i=0}^{n-1} \xi_i (W_{t_{i+1}} - W_{t_i}).}$$

Admissible class and extension. The standard L^2 theory assumes

$$\phi \text{ predictable}, \qquad \boxed{\mathbb{E} \int_0^T \phi_t^2 \, \mathrm{d}t < \infty.}$$

Choose simple predictable ϕ^n such that $\mathbb{E} \int_0^T |\phi_t^n - \phi_t|^2 \, \mathrm{d}t \to 0$. Then

$$\boxed{\int_0^T \phi_t \, \mathrm{d}W_t := L^2\text{-}\lim_{n\to\infty} \int_0^T \phi_t^n \, \mathrm{d}W_t.}$$

The Itô isometry below proves that the limit exists, is unique, and does not depend on the approximating sequence.

3.2 The identities that do the calculations

Let ϕ, ψ be predictable with finite expected energy. Then

$$\begin{aligned}
\int_0^T (a\phi_t + b\psi_t) \, \mathrm{d}W_t &= a \int_0^T \phi_t \, \mathrm{d}W_t + b \int_0^T \psi_t \, \mathrm{d}W_t, \\
\mathbb{E}\left[\int_0^T \phi_t \, \mathrm{d}W_t \right] &= 0, \\
\mathbb{E}\left[\left(\int_0^T \phi_t \, \mathrm{d}W_t \right)^2 \right] &= \mathbb{E} \int_0^T \phi_t^2 \, \mathrm{d}t, \\
\mathbb{E}\left[\left(\int_0^T \phi_t \, \mathrm{d}W_t \right) \left(\int_0^T \psi_t \, \mathrm{d}W_t \right) \right] &= \mathbb{E} \int_0^T \phi_t \psi_t \, \mathrm{d}t.
\end{aligned}$$

The third line is the *Itô isometry*; the fourth is its polarization or covariance identity. Thus integrand energy becomes terminal variance.

Theorem 3.2 (Martingale gain process)**.** *If ϕ is predictable and $\mathbb{E} \int_0^T \phi_s^2 \, \mathrm{d}s < \infty$, then*

$$M_t = \int_0^t \phi_s \, \mathrm{d}W_s$$

is a continuous square-integrable martingale. In particular, for $s \le t$,

$$\boxed{\mathbb{E}[M_t \mid \mathcal{F}_s] = M_s, \qquad \mathbb{E}\left[\int_s^t \phi_u \, \mathrm{d}W_u \middle| \mathcal{F}_s \right] = 0.}$$

The quadratic variation and mutual covariation are

$$\boxed{[M]_t = \int_0^t \phi_s^2 \, \mathrm{d}s, \qquad \left[\int_0^{\cdot} \phi_s \, \mathrm{d}W_s, \int_0^{\cdot} \psi_s \, \mathrm{d}W_s\right]_t = \int_0^t \phi_s \psi_s \, \mathrm{d}s.}$$

The first formula is pathwise; the isometry is its expected-value analogue.

Remark 3.3 (True versus local martingale). The expected condition $\mathbb{E}\int_0^T \phi_s^2 \, \mathrm{d}s < \infty$ gives a true L^2-martingale. Pathwise local square integrability generally gives only a local martingale; zero drift alone does not justify erasing the stochastic integral after taking expectations.

3.3 Deterministic integrands and formulas to know

If $f \in L^2(0,T)$ is deterministic, then

$$\boxed{\int_0^T f(t) \, \mathrm{d}W_t \sim \mathcal{N}\left(0, \int_0^T f(t)^2 \, \mathrm{d}t\right).}$$

For deterministic f, g, the two integrals are jointly Gaussian and

$$\mathrm{Cov}\left(\int_0^T f \, \mathrm{d}W, \int_0^T g \, \mathrm{d}W\right) = \int_0^T f(t)g(t) \, \mathrm{d}t.$$

Consequently, orthogonality in $L^2(0,T)$ implies independence of the two Gaussian integrals. This implication is special to deterministic integrands; a random predictable integrand need not produce a Gaussian integral.

Three reusable identities are

$$\boxed{\int_0^T c \, \mathrm{d}W_t = cW_T, \qquad \int_0^T t \, \mathrm{d}W_t \sim \mathcal{N}\left(0, \frac{T^3}{3}\right), \qquad \int_0^T W_t \, \mathrm{d}W_t = \frac{1}{2}(W_T^2 - T).}$$

For deterministic absolutely continuous f, stochastic integration by parts reduces to

$$\boxed{\int_0^T f(t) \, \mathrm{d}W_t = f(T)W_T - \int_0^T W_t f'(t) \, \mathrm{d}t.}$$

There is no covariation correction because a deterministic finite-variation function has zero covariation with Brownian motion.

If τ is a stopping time, then on every bounded horizon

$$\boxed{\int_0^t \mathbf{1}_{\{s \le \tau\}} \, \mathrm{d}W_s = W_{t\wedge\tau}, \qquad \mathbb{E}[W_{t\wedge\tau}^2] = \mathbb{E}[t \wedge \tau].}$$

Reusable calculation pattern. Before computing an integral, check in order:

1. predictability: is the coefficient known before the increment?
2. finite expected energy: is $\mathbb{E}\int_0^T \phi_t^2 \, \mathrm{d}t < \infty$?
3. structure: is the integrand deterministic, stopped, or simple?
4. target: use the isometry for second moments, covariance for joint risks, and the martingale property for conditional means.

Lecture 3 checklist

You should be able to:

- define an Itô integral first for a simple predictable process;
- check predictability and the condition $\mathbb{E}\int_0^T \phi_t^2\,\mathrm{d}t < \infty$;
- use linearity, zero mean, the Itô isometry, and the covariance identity;
- explain why $\int_0^t \phi_s\,\mathrm{d}W_s$ is a martingale;
- identify the normal law of an integral with deterministic integrand;
- compute brackets of Itô integrals;
- use deterministic integration by parts and derive $\int_0^T W_t\,\mathrm{d}W_t = \frac{1}{2}(W_T^2 - T)$;
- simulate an Itô integral with left-point Brownian increments and validate it with the isometry.

4 Lecture 4 — Itô Processes and Itô's Formula

Summary. Quadratic variation makes stochastic calculus second order, and covariation turns correlation into a $\mathrm{d}t$ term. Itô's formula is a chain rule plus a curvature correction. Its product and ratio forms solve SDEs, support discounting and delta hedging, and distinguish local from true martingales.

Differential versus integral form. For an Itô process, the compact notation and its precise meaning are

$$\boxed{\mathrm{d}X_t = a_t\,\mathrm{d}t + b_t\,\mathrm{d}W_t \quad\Longleftrightarrow\quad X_t - X_s = \int_s^t a_u\,\mathrm{d}u + \int_s^t b_u\,\mathrm{d}W_u, \qquad 0 \le s \le t.}$$

Use the differential form for calculations and the integral form as the rigorous equality. Never divide through by $\mathrm{d}t$: Brownian paths are not differentiable.

4.1 Quadratic covariation and the differential table

For continuous processes X, Y, quadratic covariation is the limiting sum

$$[X,Y]_t = \lim_{|\pi|\to 0} \sum_i (X_{t_{i+1}} - X_{t_i})(Y_{t_{i+1}} - Y_{t_i}),$$

whenever the limit exists in probability. For Brownian motions W, B with instantaneous correlation ρ,

$$\boxed{(\mathrm{d}W_t)^2 = \mathrm{d}t, \quad \mathrm{d}W_t\,\mathrm{d}B_t = \rho\,\mathrm{d}t, \quad \mathrm{d}t\,\mathrm{d}W_t = 0, \quad (\mathrm{d}t)^2 = 0.}$$

These are mnemonics for quadratic-variation limits, not ordinary algebraic infinitesimals. If

$$\mathrm{d}X_t = a_t\,\mathrm{d}t + b_t\,\mathrm{d}W_t + c_t\,\mathrm{d}B_t, \quad \mathrm{d}Y_t = p_t\,\mathrm{d}t + q_t\,\mathrm{d}W_t + r_t\,\mathrm{d}B_t,$$

then

$$\boxed{\mathrm{d}[X,Y]_t = \big(b_t q_t + c_t r_t + \rho b_t r_t + \rho c_t q_t\big)\,\mathrm{d}t.}$$

Only diffusion coefficients enter; finite-variation drifts contribute no quadratic covariation.

For the Itô process $\mathrm{d}X_t = a_t\,\mathrm{d}t + b_t\,\mathrm{d}W_t$,

$$\boxed{\mathrm{d}[X]_t = b_t^2\,\mathrm{d}t, \qquad [X]_t = \int_0^t b_s^2\,\mathrm{d}s.}$$

4.2 One- and two-dimensional Itô formulas

Theorem 4.1 (One-dimensional Itô formula)**.** *Let* $\mathrm{d}X_t = a_t\,\mathrm{d}t + b_t\,\mathrm{d}W_t$ *and* $f \in C^{1,2}$*. Then*

$$\boxed{\mathrm{d}f(t,X_t) = \left(f_t + a_t f_x + \frac{1}{2}b_t^2 f_{xx}\right)(t,X_t)\,\mathrm{d}t + b_t f_x(t,X_t)\,\mathrm{d}W_t.}$$

Equivalently, $\mathrm{d}f = f_t\,\mathrm{d}t + f_x\,\mathrm{d}X + \frac{1}{2}f_{xx}\,\mathrm{d}[X]$.

The drift vanishes when

$$\boxed{f_t + a_t f_x + \tfrac{1}{2}b_t^2 f_{xx} = 0}$$

along the process, making $f(t,X_t)$ a local martingale. A separate boundedness or integrability argument is required to obtain a true martingale and use conditional expectations.

Theorem 4.2 (Two-dimensional Itô formula)**.** *Let* X, Y *be continuous Itô processes and let* $f \in C^{1,2}([0,T]\times\mathbb{R}^2)$*, with one continuous time derivative and continuous spatial derivatives up to order two. Then*

$$\boxed{\begin{aligned}\mathrm{d}f(t,X_t,Y_t) = {} & f_t\,\mathrm{d}t + f_x\,\mathrm{d}X_t + f_y\,\mathrm{d}Y_t \\ & + \frac{1}{2}f_{xx}\,\mathrm{d}[X]_t + f_{xy}\,\mathrm{d}[X,Y]_t + \frac{1}{2}f_{yy}\,\mathrm{d}[Y]_t.\end{aligned}}$$

All derivatives are evaluated at (t,X_t,Y_t)*. In particular, if*

$$\mathrm{d}X_t = a_t\,\mathrm{d}t + b_t\,\mathrm{d}W_t, \qquad \mathrm{d}Y_t = c_t\,\mathrm{d}t + d_t\,\mathrm{d}B_t, \qquad \mathrm{d}W_t\,\mathrm{d}B_t = \rho\,\mathrm{d}t,$$

then

$$\boxed{\begin{aligned}\mathrm{d}f(t,X_t,Y_t) = {} & \left(f_t + a_t f_x + c_t f_y + \tfrac{1}{2}b_t^2 f_{xx} + \rho b_t d_t f_{xy} + \tfrac{1}{2}d_t^2 f_{yy}\right)\mathrm{d}t \\ & + b_t f_x\,\mathrm{d}W_t + d_t f_y\,\mathrm{d}B_t.\end{aligned}}$$

The mixed derivative has no factor $1/2$*; its contribution vanishes for independent Brownian drivers.*

Reliable application method. Write the state variables as placeholders, compute all first derivatives and all second derivatives in noisy directions, substitute the processes, then group separately the $\mathrm{d}t$ and Brownian terms. Always check the domain of f, especially for $\log x$, $1/x$, and payoff kinks.

4.3 Reusable transformation rules

If $X_t > 0$ and $\mathrm{d}X_t/X_t = \alpha_t\,\mathrm{d}t + \beta_t\,\mathrm{d}W_t$, then

$$\boxed{\mathrm{d}\log X_t = \left(\alpha_t - \frac{1}{2}\beta_t^2\right)\mathrm{d}t + \beta_t\,\mathrm{d}W_t.}$$

The term $-\beta_t^2/2$ is the volatility correction. If $\mathrm{d}X_t = a_t\,\mathrm{d}t + b_t\,\mathrm{d}W_t$ and $X_t \neq 0$, then

$$\boxed{\mathrm{d}\left(\frac{1}{X_t}\right) = \left(-\frac{a_t}{X_t^2} + \frac{b_t^2}{X_t^3}\right)\mathrm{d}t - \frac{b_t}{X_t^2}\,\mathrm{d}W_t.}$$

For continuous Itô processes,

$$\boxed{\mathrm{d}(X_tY_t) = X_t\,\mathrm{d}Y_t + Y_t\,\mathrm{d}X_t + \mathrm{d}[X,Y]_t.}$$

4.4 Canonical SDEs to solve and reuse

Geometric Brownian motion. For

$$\mathrm{d}S_t = \mu S_t\,\mathrm{d}t + \sigma S_t\,\mathrm{d}W_t, \qquad S_0 > 0,$$

apply Itô to $\log S_t$:

$$\boxed{S_t = S_0 \exp\left\{\left(\mu - \frac{1}{2}\sigma^2\right)t + \sigma W_t\right\}.}$$

Thus $S_t > 0$, $\log S_t$ is Gaussian, and

$$\mathbb{E}[S_t] = S_0 e^{\mu t}, \qquad \frac{\mathrm{d}(S_t^p)}{S_t^p} = \left[p\mu + \frac{1}{2}p(p-1)\sigma^2\right]\mathrm{d}t + p\sigma\,\mathrm{d}W_t.$$

Ornstein–Uhlenbeck process. For

$$\mathrm{d}X_t = \kappa(m - X_t)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t, \qquad \kappa > 0,$$

the integrating factor $e^{\kappa t}$ yields, for $s \le t$,

$$\boxed{X_t = m + (X_s - m)e^{-\kappa(t-s)} + \sigma\int_s^t e^{-\kappa(t-u)}\,\mathrm{d}W_u.}$$

Therefore

$$\boxed{\mathbb{E}[X_t \mid \mathcal{F}_s] = m + (X_s - m)e^{-\kappa(t-s)}, \quad \mathrm{Var}(X_t \mid \mathcal{F}_s) = \frac{\sigma^2}{2\kappa}\big(1 - e^{-2\kappa(t-s)}\big).}$$

Exponential martingale and Brownian moments. Two particularly useful drift cancellations are

$$\boxed{\exp\left(\lambda W_t - \frac{1}{2}\lambda^2 t\right) \text{ is a martingale}, \qquad W_t^2 - t \text{ is a martingale.}}$$

More generally, applying Itô's formula to W_t^n gives the recursion

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}[W_t^n] = \frac{n(n-1)}{2}\mathbb{E}[W_t^{n-2}],$$

whenever the stochastic integral is integrable.

4.5 Several factors and instantaneous hedging

If the unhedged shock is $\sigma_X\,\mathrm{d}W_t - h\sigma_Y\,\mathrm{d}B_t$, with $\mathrm{d}W_t\,\mathrm{d}B_t = \rho\,\mathrm{d}t$, its variance rate is

$$q(h) = \sigma_X^2 + h^2\sigma_Y^2 - 2h\rho\sigma_X\sigma_Y.$$

The minimum-variance hedge is

$$\boxed{h^* = \rho\frac{\sigma_X}{\sigma_Y}, \qquad q(h^*) = \sigma_X^2(1 - \rho^2).}$$

Only the component correlated with the hedging asset can be removed.

Lecture 4 checklist

You should be able to:

- translate an Itô process between differential and integral forms;
- identify the time and state variables and write every driving SDE;
- compute f_t, f_x, and f_{xx} before applying Itô's formula;
- apply the two-dimensional formula, including f_y, f_{yy}, and the mixed term $f_{xy}\,\mathrm{d}[X,Y]_t$;
- calculate covariations, including $\mathrm{d}W_t\,\mathrm{d}B_t = \rho\,\mathrm{d}t$ for correlated drivers;
- retain all terms of order $\mathrm{d}t$ and separate the final drift and diffusion terms;
- use product, quotient, and integration-factor formulas for Itô processes;
- solve standard SDEs such as GBM and Ornstein–Uhlenbeck processes;
- check the domain, $C^{1,2}$ regularity, and integrability conditions;
- state the probability measure and financing convention in a financial application.

5 Lecture 5 — Self-Financing, Replication, and Black–Scholes

Summary. Itô's formula identifies the dynamics of a candidate price. Self-financing specifies how a trading strategy earns its gains. Matching those gains with a European claim gives the Black–Scholes PDE and its delta hedge. The Gaussian solution produces the call formula and its Greeks. The basic model in this lecture has no dividends.

5.1 From Itô's formula to a pricing equation

For a stock with historical dynamics

$$\mathrm{d}S_t = \mu S_t\,\mathrm{d}t + \sigma S_t\,\mathrm{d}W_t^{\mathbb{P}}, \qquad \mathrm{d}B_t = rB_t\,\mathrm{d}t, \qquad B_0 = 1,$$

Itô's formula applied directly to $V_t = v(t,S_t)$ gives

$$\mathrm{d}V_t = \left(v_t + \mu S_t v_s + \tfrac{1}{2}\sigma^2 S_t^2 v_{ss}\right)\,\mathrm{d}t + \sigma S_t v_s\,\mathrm{d}W_t^{\mathbb{P}}.$$

The second derivative term comes from $\mathrm{d}[S]_t = \sigma^2 S_t^2\,\mathrm{d}t$. For path-dependent payoffs, a running integral such as $A_t = \int_0^t S_u\,\mathrm{d}u$ can be added to the state. This idea returns in Lecture 7.

5.2 The self-financing budget

Let Δ_t be stock shares and β_t bank-account units. The portfolio value is $X_t = \Delta_t S_t + \beta_t B_t$. If holdings change at a trading date t_i, the new shares must be paid for from the bank account:

$$S_{t_i}(\Delta_i - \Delta_{i-1}) + B_{t_i}(\beta_i - \beta_{i-1}) = 0.$$

The holdings chosen before the next price move earn that move. Passing from these left-point gains to continuous time gives the self-financing condition

$$\boxed{X_t = X_0 + \int_0^t \Delta_u\,\mathrm{d}S_u + \int_0^t \beta_u\,\mathrm{d}B_u, \qquad \mathrm{d}X_t = \Delta_t\,\mathrm{d}S_t + r(X_t - \Delta_t S_t)\,\mathrm{d}t.}$$

The value identity $X = \Delta S + \beta B$ by itself does not impose self-financing. For continuous finite-variation holdings, the product rule also gives $S_t\,\mathrm{d}\Delta_t + B_t\,\mathrm{d}\beta_t = 0$. For more general semimartingale holdings, a covariation term must be included in that rebalancing identity; the stochastic-integral gain equation remains the definition.

5.3 Replication, the PDE, and verification

Suppose a European claim pays $g(S_T)$ and has smooth candidate price $v(t, S_t)$, with $v(T, s) = g(s)$. A replicating portfolio with $X_t = v(t, S_t)$ has gain

$$\mathrm{d}X_t = [rv + (\mu - r)S_t\Delta_t]\,\mathrm{d}t + \sigma S_t\Delta_t\,\mathrm{d}W_t^{\mathbb{P}}.$$

Matching its Brownian coefficient with that of $\mathrm{d}v(t, S_t)$ gives $\Delta_t = v_s(t, S_t)$. Matching drifts then cancels μ and yields

$$\boxed{v_t + rs\,v_s + \tfrac{1}{2}\sigma^2 s^2 v_{ss} - rv = 0, \qquad v(T, s) = g(s).}$$

The cash holding is

$$\boxed{\beta_t = \frac{v(t, S_t) - S_t v_s(t, S_t)}{B_t}.}$$

Conversely, a sufficiently regular PDE solution with an admissible self-financing strategy and the correct terminal payoff verifies the replication. No arbitrage then identifies its initial cost as the claim price. The argument uses one traded Brownian exposure to span the one modeled Brownian risk.

5.4 Gaussian solution, call formula, and Greeks

Set $\tau = T - t$. The logarithmic change of variable and removal of the discounting term transform the PDE into a heat equation. Its Gaussian representation is

$$v(t, s) = e^{-r\tau}\mathbb{E}\Big[g\Big(se^{(r-\sigma^2/2)\tau + \sigma\sqrt{\tau}Z}\Big)\Big], \qquad Z \sim N(0, 1).$$

For a call with strike $K > 0$, put

$$d_1 = \frac{\log(s/K) + (r + \sigma^2/2)\tau}{\sigma\sqrt{\tau}}, \qquad d_2 = d_1 - \sigma\sqrt{\tau}.$$

The call, put, and parity formulas are

$$\boxed{\begin{aligned} C(t, s) &= s\Phi(d_1) - Ke^{-r\tau}\Phi(d_2), \\ P(t, s) &= Ke^{-r\tau}\Phi(-d_2) - s\Phi(-d_1), \\ C(t, s) - P(t, s) &= s - Ke^{-r\tau}. \end{aligned}}$$

Here Φ and $\varphi = \Phi'$ are the standard normal CDF and density. The identity

$$\boxed{s\varphi(d_1) = Ke^{-r\tau}\varphi(d_2)}$$

cancels the density terms when differentiating. With calendar-time $\Theta = \partial_t C$, the call Greeks are

$$\boxed{\begin{aligned} &\Delta_C = \Phi(d_1), && \Gamma_C = \frac{\varphi(d_1)}{s\sigma\sqrt{\tau}}, && \mathrm{Vega}_C = s\varphi(d_1)\sqrt{\tau}, \\ &\Theta_C = -\frac{s\sigma\varphi(d_1)}{2\sqrt{\tau}} - rKe^{-r\tau}\Phi(d_2), && \rho_C = K\tau e^{-r\tau}\Phi(d_2). \end{aligned}}$$

Delta is the stock holding in the hedge. Gamma measures how quickly that holding changes. Discrete rebalancing, funding frictions, jumps, and volatility misspecification make the continuous-time hedge imperfect.

Lecture 5 checklist

- distinguish portfolio value from self-financing gains;
- match diffusion and drift to obtain delta and the pricing PDE;
- recover the call formula and differentiate it to obtain the Greeks;
- check regularity, integrability, and the terminal payoff when verifying a candidate hedge.

6 Lecture 6 — Reweighting, Girsanov, and Risk-Neutral Measure

Summary. A change of measure visits events with different frequencies while an inverse density preserves the original expectation. Gaussian tilting becomes Girsanov's drift shift for Brownian paths. A true density martingale is needed before the risk-neutral measure can be used.

6.1 Rare tails motivate a change of weights

A change of measure can make a rare event easier to sample. To recover its original probability, reweight the observations by the inverse density:

$$\mathbb{P}(A) = \mathbb{E}^{\mathbb{Q}}\left[\frac{\mathrm{d}\mathbb{P}}{\mathrm{d}\mathbb{Q}}\mathbf{1}_A\right]$$

for equivalent measures $\mathbb{P}$ and $\mathbb{Q}$.

6.2 Densities and conditional Bayes' formula

If $Z_T > 0$, $\mathbb{E}^{\mathbb{P}}[Z_T] = 1$, and $\mathrm{d}\mathbb{Q}/\,\mathrm{d}\mathbb{P}|_{\mathcal{F}_T} = Z_T$, then for integrable Y,

$$\boxed{\mathbb{E}^{\mathbb{Q}}[Y] = \mathbb{E}^{\mathbb{P}}[Z_T Y].}$$

The density process is $Z_t = \mathbb{E}^{\mathbb{P}}[Z_T \mid \mathcal{F}_t]$. For $\mathcal{G} \subseteq \mathcal{F}_T$, the conditional version is

$$\boxed{\mathbb{E}^{\mathbb{Q}}[Y \mid \mathcal{G}] = \frac{\mathbb{E}^{\mathbb{P}}[Z_T Y \mid \mathcal{G}]}{\mathbb{E}^{\mathbb{P}}[Z_T \mid \mathcal{G}]}.}$$

Changing measure changes the weights of paths, while equivalent measures agree on null events.

6.3 Stochastic exponential and the sign of Girsanov

For a predictable process λ with finite pathwise energy, the candidate density is

$$\boxed{Z_t = \exp\left(-\int_0^t \lambda_u \, \mathrm{d}W_u^{\mathbb{P}} - \tfrac{1}{2}\int_0^t \lambda_u^2 \, \mathrm{d}u\right), \qquad \mathrm{d}Z_t = -Z_t \lambda_t \, \mathrm{d}W_t^{\mathbb{P}}.}$$

This is initially a nonnegative local martingale. A sufficient condition for it to be a true martingale on $[0, T]$ is Novikov's condition

$$\mathbb{E}^{\mathbb{P}}\left[\exp\left(\tfrac{1}{2}\int_0^T \lambda_u^2 \, \mathrm{d}u\right)\right] < \infty.$$

If the density is a true martingale and $\mathrm{d}\mathbb{Q}/\,\mathrm{d}\mathbb{P}|_{\mathcal{F}_T} = Z_T$, Girsanov gives

$$\boxed{W_t^{\mathbb{Q}} = W_t^{\mathbb{P}} + \int_0^t \lambda_u \, \mathrm{d}u.}$$

Thus $\mathrm{d}W_t^{\mathbb{P}} = \mathrm{d}W_t^{\mathbb{Q}} - \lambda_t \, \mathrm{d}t$. If $\mathrm{d}X_t = b_t \, \mathrm{d}t + a_t \, \mathrm{d}W_t^{\mathbb{P}}$, then

$$\mathrm{d}X_t = (b_t - a_t\lambda_t) \, \mathrm{d}t + a_t \, \mathrm{d}W_t^{\mathbb{Q}}.$$

The drift changes; the quadratic variation $\mathrm{d}[X]_t = a_t^2 \, \mathrm{d}t$ does not.

6.4 The Black–Scholes pricing measure

For the non-dividend-paying stock in Lecture 5, set $\lambda = (\mu - r)/\sigma$. With the corresponding true density martingale,

$$\boxed{\frac{\mathrm{d}S_t}{S_t} = r \, \mathrm{d}t + \sigma \, \mathrm{d}W_t^{\mathbb{Q}}, \qquad \mathrm{d}(e^{-rt}S_t) = \sigma e^{-rt} S_t \, \mathrm{d}W_t^{\mathbb{Q}}.}$$

The discounted price is a local martingale, and in this constant coefficient model it is a true martingale. The measure $\mathbb{Q}$ is a pricing measure, not a forecast of historical returns. Its conditional lognormal law also yields $C_t = S_t\Phi(d_1) - Ke^{-r(T-t)}\Phi(d_2)$ from two Gaussian tails.

Lecture 6 checklist

- write a density and its inverse before reweighting an expectation;
- distinguish a local exponential from a true density martingale;
- keep the sign of the Brownian drift shift consistent with Z;
- derive the risk-neutral drift of a non-dividend-paying stock.

7 Lecture 7 — Martingale Prices, Hedges, and Completeness

Summary. Discounted self-financing gains lead to conditional valuation when the relevant local martingale is a true martingale. Dividends enter the stock gain and change its risk-neutral price drift. Changing numeraire changes the units and measure together. Martingale representation identifies a claim's Brownian exposure; traded assets must span that exposure to implement a hedge.

7.1 From stochastic integrals to martingale prices

Let $B_t = \exp(\int_0^t r_u \, \mathrm{d}u)$ and $D_t = B_t^{-1}$. For a non-dividend-paying stock under a money-market pricing measure,

$$\frac{\mathrm{d}S_t}{S_t} = r_t \, \mathrm{d}t + \sigma_t \, \mathrm{d}W_t^{\mathbb{Q}} \quad \Longrightarrow \quad \mathrm{d}(D_t S_t) = D_t \sigma_t S_t \, \mathrm{d}W_t^{\mathbb{Q}}.$$

The discounted price is a local martingale because it is a stochastic integral. For a self-financing portfolio $X_t = \Delta_t S_t + \beta_t B_t$, integration by parts gives

$$\boxed{\mathrm{d}(D_t X_t) = \Delta_t \, \mathrm{d}(D_t S_t).}$$

If this discounted wealth is a true martingale and $X_T = H$, then

$$\boxed{V_t = X_t = B_t \mathbb{E}^{\mathbb{Q}}\left[\frac{H}{B_T} \middle| \mathcal{F}_t \right].}$$

Local martingality alone does not justify the conditional-expectation equality; integrability or an appropriate admissibility argument is needed.

7.2 Dividend gains and the pricing equation

Suppose the stock pays continuous dividends $qS_t \, \mathrm{d}t$. Its *historical ex-dividend price* still has the form

$$\mathrm{d}S_t = \mu S_t \, \mathrm{d}t + \sigma S_t \, \mathrm{d}W_t^{\mathbb{P}}.$$

The gain from one share is $\mathrm{d}S_t + qS_t \, \mathrm{d}t$. A self-financing hedge therefore satisfies

$$\boxed{\mathrm{d}X_t = \Delta_t(\, \mathrm{d}S_t + qS_t \, \mathrm{d}t) + \beta_t \, \mathrm{d}B_t,}$$

and its discounted form is

$$\mathrm{d}(D_t X_t) = \Delta_t \left[\mathrm{d}(D_t S_t) + D_t q S_t \, \mathrm{d}t \right].$$

For $V_t = v(t, S_t)$, matching Brownian coefficients gives $\Delta_t = v_s$. Matching drifts cancels the historical price drift μ and gives

$$\boxed{v_t + (r-q)sv_s + \tfrac{1}{2}\sigma^2 s^2 v_{ss} - rv = 0.}$$

The matching Brownian shift is $\lambda = (\mu + q - r)/\sigma$, so under the associated pricing measure

$$\frac{\mathrm{d}S_t}{S_t} = (r-q) \, \mathrm{d}t + \sigma \, \mathrm{d}W_t^{\mathbb{Q}}.$$

The discounted *gain* is driftless; the discounted ex-dividend price alone has dividend-related drift.

7.3 Change of numeraire

For a strictly positive traded numeraire N with associated measure $\mathbb{Q}^N$, a replicable payoff has the representation

$$\boxed{V_t = N_t \mathbb{E}^{\mathbb{Q}^N}\left[\frac{H}{N_T} \middle| \mathcal{F}_t \right].}$$

The numeraire, its measure, and the ratio H/N_T form one matched choice of units. Taking $N = B$ recovers money-market discounting.

7.4 Martingale representation turns prices into hedges

In a Brownian filtration, a square-integrable discounted claim value has a representation

$$M_t = \mathbb{E}^{\mathbb{Q}}[D_T H \mid \mathcal{F}_t] = M_0 + \int_0^t \psi_u^\top \, \mathrm{d}W_u^{\mathbb{Q}}.$$

The vector ψ_t is the claim's exposure to Brownian shocks. With one non-dividend-paying traded stock satisfying $\mathrm{d}(D_t S_t) = D_t \sigma_t S_t \, \mathrm{d}W_t^{\mathbb{Q}}$, exposure matching gives

$$\boxed{\Delta_t = \frac{\psi_t}{D_t \sigma_t S_t}.}$$

If the claim is $v(t, S_t)$, this reduces to $\Delta_t = v_s(t, S_t)$. For m assets driven by d Brownian factors, write $\mathrm{d}\bar{S}_t = \Gamma_t \, \mathrm{d}W_t^{\mathbb{Q}}$, with $\Gamma_t \in \mathbb{R}^{m \times d}$. Holdings ϕ_t implement a claim exposure ψ_t when

$$\boxed{\Gamma_t^\top \phi_t = \psi_t.}$$

7.5 Completeness and the spanning condition

In a d-factor Brownian market, all Brownian exposures can be spanned when the discounted volatility matrix satisfies

$$\boxed{\operatorname{rank}(\Gamma_t) = d \quad \text{for almost every } (t, \omega).}$$

Together with the required admissibility and integrability conditions, this is the completeness criterion. A new traded asset helps only if its Brownian exposure adds a missing direction.

When a Brownian direction is unspanned, its risk-premium shift can remain free across equivalent martingale measures. Claims exposed to that direction need not have a unique no-arbitrage price.

7.6 Path dependence and the choice of state

The full-filtration pricing identity applies to path-dependent claims. To express the price as a function of the current state, include enough variables to record the relevant path information. Apply the multidimensional Itô formula to this enlarged state: a finite-variation state contributes a first derivative but no quadratic-variation or covariation term.

Lecture 7 checklist

- distinguish discounted prices from discounted dividend gains;
- derive the dividend-adjusted PDE and delta hedge by replication;
- match a numeraire with its measure in a valuation formula;
- use martingale representation and a rank test to identify hedgeable Brownian exposures;
- identify the additional state variables needed for a path-dependent claim.

Examples to know.

- Price a unit payoff, a terminal share, a forward, and a digital; recover a European call by decomposing its payoff.
- Price and hedge a call with continuous dividends; recover put–call parity using a prepaid share.

- Choose a numeraire that simplifies the terminal payoff ratio.
- Find the martingale representations of linear and quadratic Brownian terminal payoffs.
- Analyze a stock driven by two Brownian risks and a claim depending on an untraded sector state; determine whether a second traded product completes the market.
- Use a nontraded claim to illustrate price ambiguity in an incomplete market.
- Introduce the running average for an arithmetic Asian option and derive its two-state pricing PDE.